

INTRODUCTION

Modern networked systems ranging from power grids to multi-robot fleets to neuronal brain networks pose unique challenges due to their large size and complexity of state interactions. To confront these challenges, our *ultimate goal is to design sparse optimal controllers for networked systems with multiplicative noise from state trajectory data*. Our current work builds toward this goal in 4 steps:

1. Non-sparse optimal control when model is known [well-known in the literature]
2. Non-sparse optimal control when model is unknown [1]
3. Sparse optimal control when model is known [2]
4. Sparse optimal control when model is unknown [forthcoming]

Typical optimal control results are valid only for linear models which are deterministic or have only additive noise. However, multiplicative noise is a more natural modeling mechanism for capturing noise (sensor noise, process noise, nonlinearities and other unmodeled dynamics, etc.) on edges of networked systems e.g. due to electrical noise on power lines, intermittent communication losses between multiple robots or airborne vehicles, aerodynamic interactions between rotorcraft or wind turbines in close proximity, etc. Designing controllers which account for this type of stochastic dynamics ensures that the closed-loop system satisfy a generalized and necessary notion of stability called *mean-square stability* based on second moments (state covariance).

Definition: A system is called *mean-square stable*, or second moment bounded, if there exists a positive constant α such that all state trajectories $x(t)$ satisfy

$$\lim_{t \rightarrow \infty} \mathbb{E}[x(t)x(t)^T] \leq \alpha, \quad \forall x(0) \in \mathbb{R}^n$$

Importantly, failing to account for this noise in the control design can lead the closed-loop system to become mean-square unstable *even if the open-loop system was mean-square stable!*

Consider the optimal control problem for discrete-time linear time-invariant systems with multiplicative noise and quadratic costs:

$$\min_{\pi \in \Pi} J(\pi) = \mathbb{E} \sum_{t=0}^{\infty} (x_t^T Q x_t + u_t^T R u_t),$$

$$\text{s.t. } x_{t+1} = (A + \sum_{i=1}^p \delta_{it} A_i) x_t + (B + \sum_{j=1}^q \gamma_{jt} B_j) u_t$$

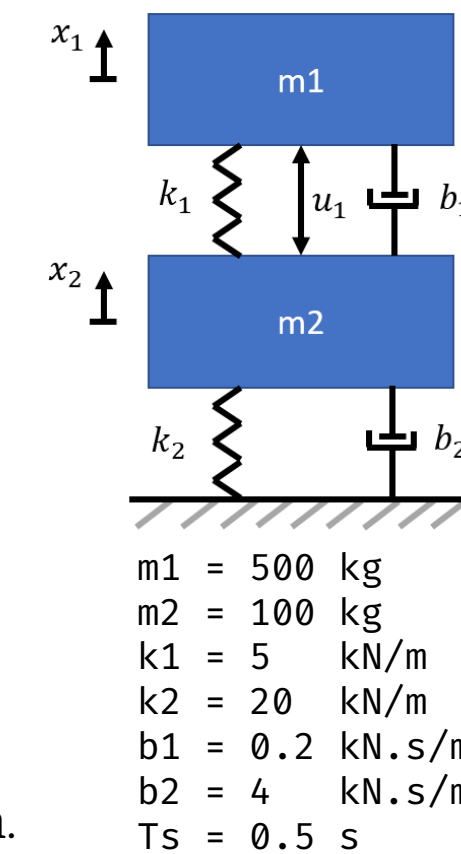
The solution is linear state feedback, found from generalized algebraic Riccati equations.

$$P = Q + A^T P A + \sum_{i=1}^p \alpha_i A_i^T P A_i - A^T P B (R + B^T P B + \sum_{j=1}^q \beta_j B_j^T P B_j)^{-1} B^T P A$$

$$K^* = -(R + B^T P B + \sum_{j=1}^q \beta_j B_j^T P B_j)^{-1} B^T P A$$

Example: Two mass suspension

LQR cost	Noise-aware control	Noise-ignorant control
System has some true noise	483.9	∞
System has zero true noise	27.5	26.3
Gain matrix	$\begin{bmatrix} 2.2633 & -0.0378 \\ -0.0397 & 0.0038 \end{bmatrix}$	$\begin{bmatrix} 2.6231 & 0.0635 \\ -0.0010 & 0.0000 \end{bmatrix}$



Ignoring multiplicative noise when present in the true dynamics is disastrous - the control leads to instability! This happens despite superficial similarity between the controls. Conversely, assuming multiplicative noise not in the true dynamics only degrades performance slightly, does not destabilize the system.

LEARNING OPTIMAL CONTROL FROM DATA

Following the results of [3], we have shown recently in [1] that the reinforcement learning *policy gradient algorithm globally converges to the global optimal control at a linear (geometric) rate*. Policy gradient proceeds by starting with a mean-square stabilizing controller, calculating the gradient of the cost with respect to the gains, taking a (small) step in that direction, and iterating until convergence. In particular, the gradient of the objective is

$$\nabla_K J(K) = 2 \left[(R + B^T P_K B + \sum_{j=1}^q \beta_j B_j^T P_K B_j) K + B^T P_K \Sigma_K \right]$$

By leveraging *gradient domination* / *Polyak-Łojasiewicz property* of the objective, chains of matrix inequalities, and modern perturbation and concentration bounds for random matrices e.g. those of Weyl and Bernstein, we obtain a global convergence result where importantly the maximum *step size is a polynomial in the problem data*.

Theorem 3.6 (Policy gradient convergence). *Using the policy gradient step*

$$K_{s+1} = K_s - \eta \nabla_K C(K_s)$$

with step size $0 < \eta \leq c_{pg}$ gives global convergence to the optimal gain matrix K^ at the linear rate*

$$\left(C(K_{s+1}) - C(K^*) \right) \leq \left(1 - 2\eta \frac{\sigma_{\min}(R) \sigma_{\min}(\Sigma_0)^2}{\|\Sigma_{K^*}\|} \right) \left(C(K_s) - C(K^*) \right)$$

In the model-free setting, the gradient is instead estimated from sample trajectories:

Algorithm 1: Model-Free policy gradient estimation

Input: Gain matrix K , number of samples n_{sample} , rollout length ℓ , exploration radius r .

for $i = 1, \dots, n_{\text{sample}}$ **do**
 Generate a sample gain matrix $\hat{K}_i = K + U_i$, where U_i is drawn uniformly at random over matrices with Frobenius norm r .
 Generate a sample initial state $x_0^{(i)} \sim \mathcal{P}_0$.
 Simulate the closed-loop system for ℓ steps starting from $x_0^{(i)}$ yielding the state sequence $\{x_t^{(i)}\}$.
 Collect the finite-horizon cost estimate $\hat{C}_i = \sum_{t=0}^{\ell} x_t^{(i)T} (Q + \hat{K}_i^T R \hat{K}_i) x_t^{(i)}$
end
Output: Gradient estimate $\hat{\nabla} C(K) = \frac{1}{n_{\text{sample}}} \sum_{i=1}^{n_{\text{sample}}} \frac{\partial \hat{C}_i}{\partial K} U_i$.

Theorem 4.1 (Model-Free Policy Gradient convergence). *Suppose the step size η is chosen according to the restriction in Theorem 3.6 and at every iteration the gradient is estimated using Algorithm 1 where the number of samples n_{sample} , rollout length ℓ , and exploration radius r are chosen according to fixed quantities $h_{tr, trunc, GD}, h_{sample, trunc, GD}, h_{tr, trunc, GD}$ which are polynomial in the problem data $A, B, \alpha_i, \beta_j, A_i, B_j, Q, R, \Sigma_0, C(K_0)$. Then with high probability of at least $1 - \exp(-mn)$ performing gradient descent results in convergence to the global optimum at the linear rate*

$$\left(C(K_{s+1}) - C(K^*) \right) \leq \left(1 - 2\eta \frac{\sigma_{\min}(R) \sigma_{\min}(\Sigma_0)^2}{\|\Sigma_{K^*}\|} \right) \left(C(K_s) - C(K^*) \right).$$

SPARSE OPTIMAL CONTROL

Practical network controllers are made *less complex and costly* by reducing actuators, sensors, and sensor-actuator links. Building on this idea in [2], we proposed the addition of regularization to promote sparsity on the gain matrix K , leading to the sparse controller structure problem:

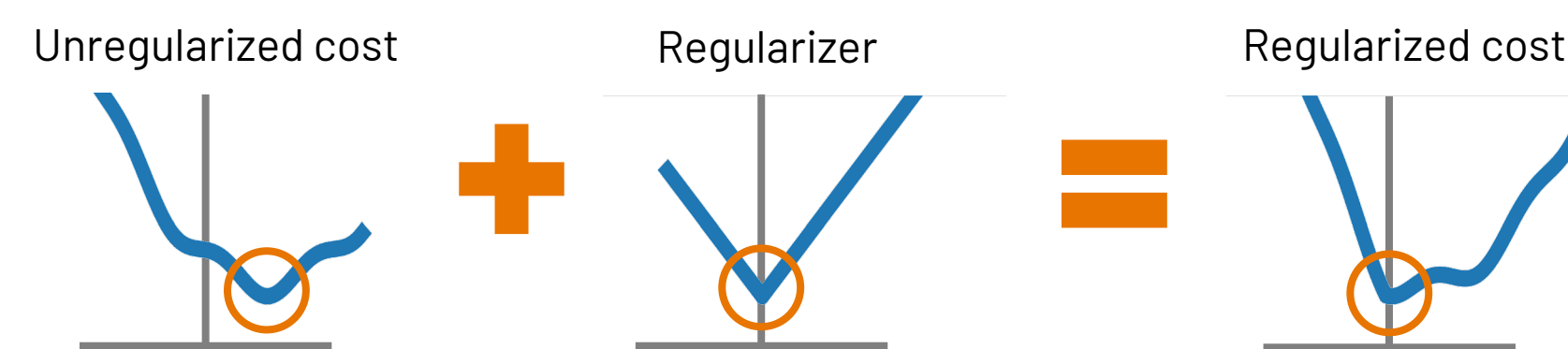
$$\min_{\pi \in \Pi} J(\pi) + \gamma J_{\text{reg}}(\pi)$$

Although the problem is nonconvex in K and may have multiple minima due to regularization, empirically the optimization *efficiently finds stabilizing, high performance sparse controllers*. Although recent works e.g. [4] cast this problem in a convex formulation, this requires a bilinear transformation of the gains, complicating the preservation of sparsity patterns “backwards” through the transformation, particularly entrywise sparsity. In contrast, the policy gradient algorithm presented here *operates directly on the gains and all sparsity patterns remain intact*.

The set of mean-square stabilizing controllers, which is the feasible set for the optimization, is nonconvex in K . However, since the regularized problem has the same feasible set as the unregularized problem, if the initial control is stabilizing, then every succeeding improved control found during the optimization is stabilizing as well.

Although one motivation for using policy gradient here is to maintain sparsity patterns, the algorithm is readily extended to the model-unknown case by simply replacing the explicit model-based gradient calculation with a model-free, data-driven gradient estimation scheme. We plan to explore convergence results for this setting in future work.

This “cartoon” demonstrates the action of regularization on the gradient dominated cost:



Regularizers $J_{\text{reg}}(K) = \|K\|_M$ promote sparsity of entries, rows, and/or columns.

$$\|K\|_M = \sum_{i=1}^m \sum_{j=1}^n |K_{ij}| \text{ or } \sum_{i=1}^m \|K^{r,i}\|_p \text{ or } \sum_{j=1}^n \|K^{c,j}\|_p$$

where $K^{r,i}$ and $K^{c,j}$ are the i^{th} row and column of K and $p = 2$ or ∞ .

These regularizers are *nondifferentiable* so we use *three optimization methods*:

1. Subgradient descent
 $K^{(k+1)} = K^{(k)} - \eta (\nabla_K J(K^{(k)}) + \gamma g_K \|K^{(k)}\|_M)$
 $K_{\text{best}} = \arg\max \{C(K^{(k+1)}), C(K_{\text{best}})\}$
 where $C(K) = J(K) + \gamma \|K\|_M$ and g_K is a subgradient.
2. Gradient descent with Huber (smooth) regularizer approximation
 $K^{(k+1)} = K^{(k)} - \eta \nabla_K J(K^{(k)}) + \gamma \|K\|_M$
3. Proximal gradient descent
 $K_S^{(k)} = K^{(k)} - \eta \nabla_K J(K^{(k)})$
 $K^{(k+1)} = \text{prox}_{\eta \|K\|_M}(K_S^{(k)})$

Table 1: Regularizer (sub)gradients

$\ K\ _M$	$g_K \ K\ _M$	$\nabla_K \ K\ _{M,h,\phi}$
$\ K\ _{1,i,j}$	$\text{sgn}(K_{ij})$	$K_{ij} / \max(K_{ij} , \phi)$
$\ K\ _{gIr}^{r,i}$	$K^{r,i} / \ K^{r,i}\ _2$	$K^{r,i} / \max(\ K^{r,i}\ _2, \phi)$

NUMERICAL EXPERIMENTS: RANDOM NETWORKS

Numerical experiments were conducted on systems representing a diffusion process on connected, *random Erdős-Rényi graphs* with 51 nodes (51 states, 51 inputs) and 2 random multiplicative noises on the states and inputs each.

We found that, with high regularization, *a large number of actuator-sensor links (entries), actuators (rows), and sensors (columns) could be deleted without compromising stability and while maintaining usable performance*. All three methods presented all gave very similar results with similar efficacy; arbitrarily entrywise and row sparse mean-square stabilizing solutions were obtained in both low- and high-noise settings after a reasonable amount of computation time.

From a computational perspective, the main expense came from evaluating the policy gradient at each iteration which required solving a generalized discrete Lyapunov equation. The compute time increased as the regularization weight was increased and a larger number of smaller steps were required to accommodate the increasing gradient magnitude.

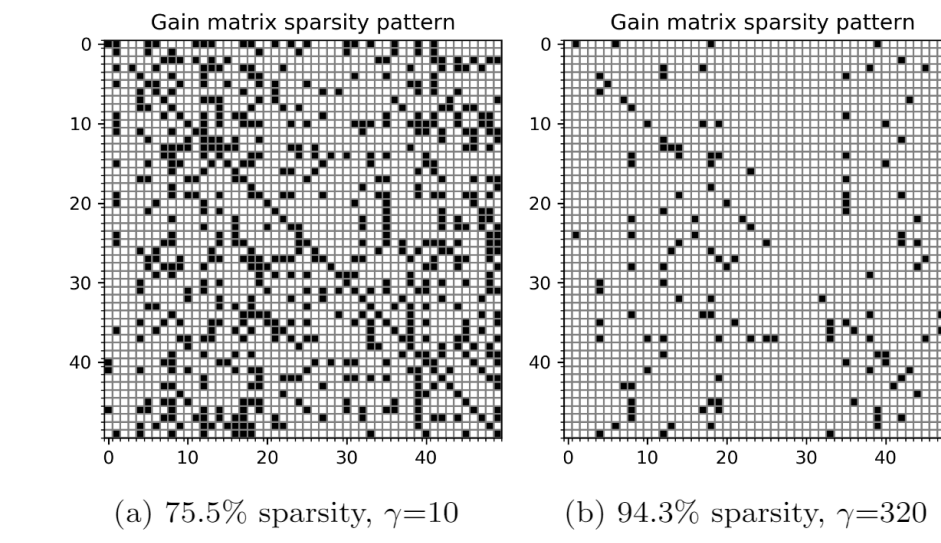
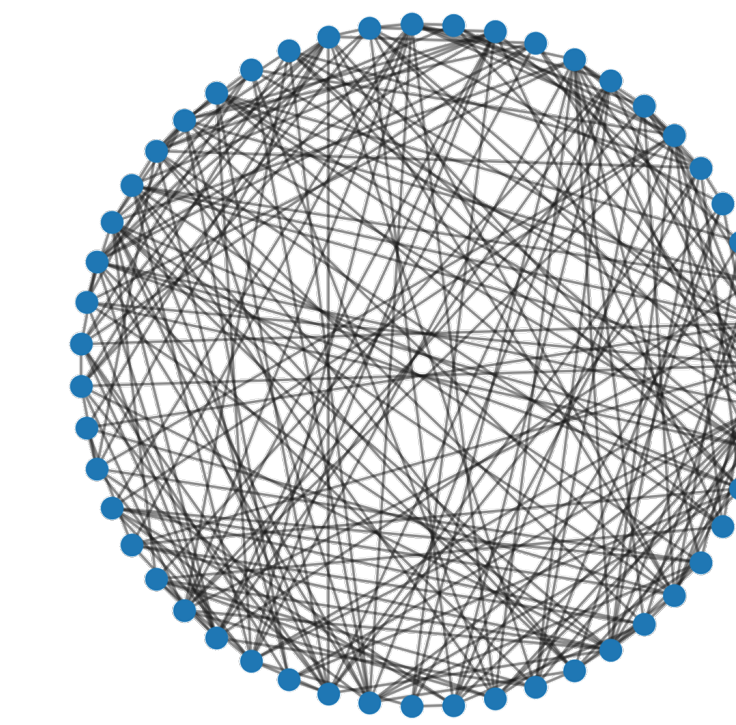


Fig. 1. Sparsity patterns for low noise, subgradient descent on the l_1 -norm regularized LQRm cost.

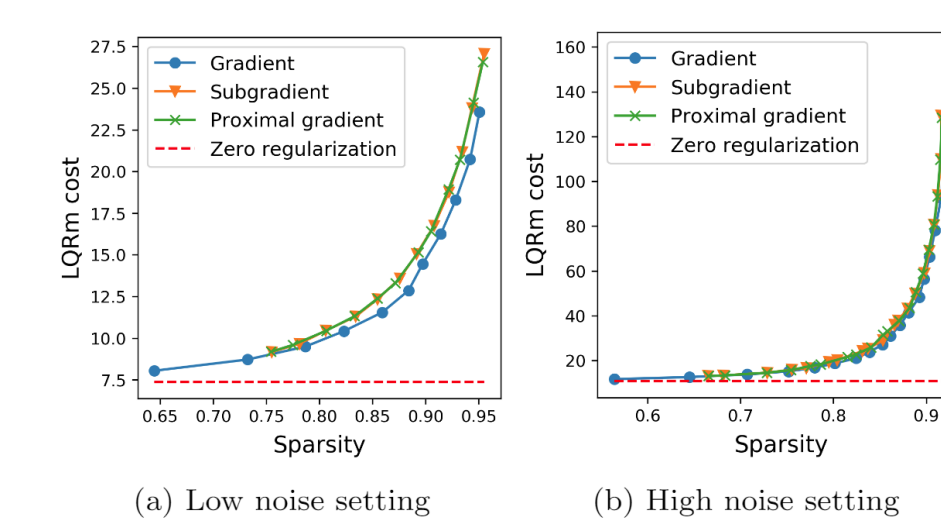


Fig. 3. LQRm cost vs. sparsity with l_1 -norm regularization.

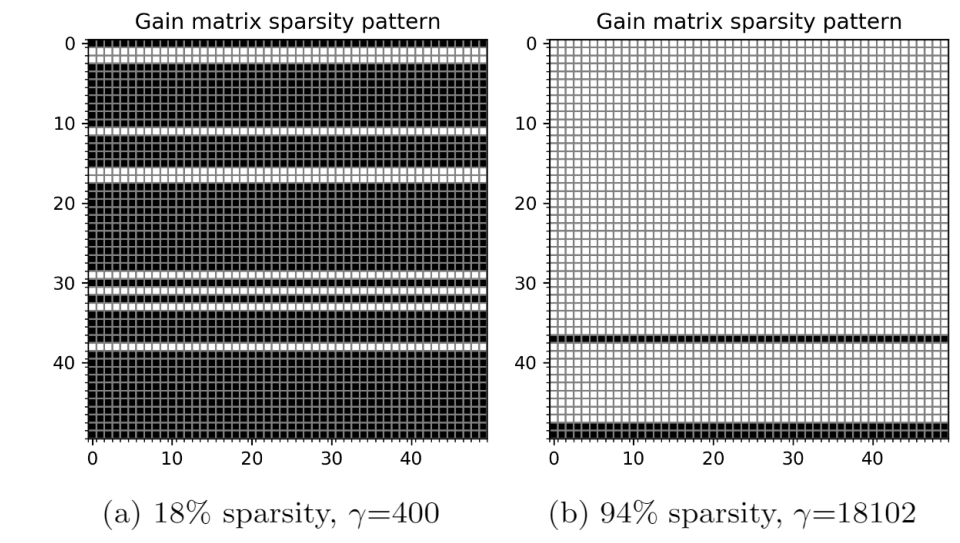


Fig. 2. Sparsity patterns for low noise, subgradient descent on the row-group LASSO regularized LQRm cost.

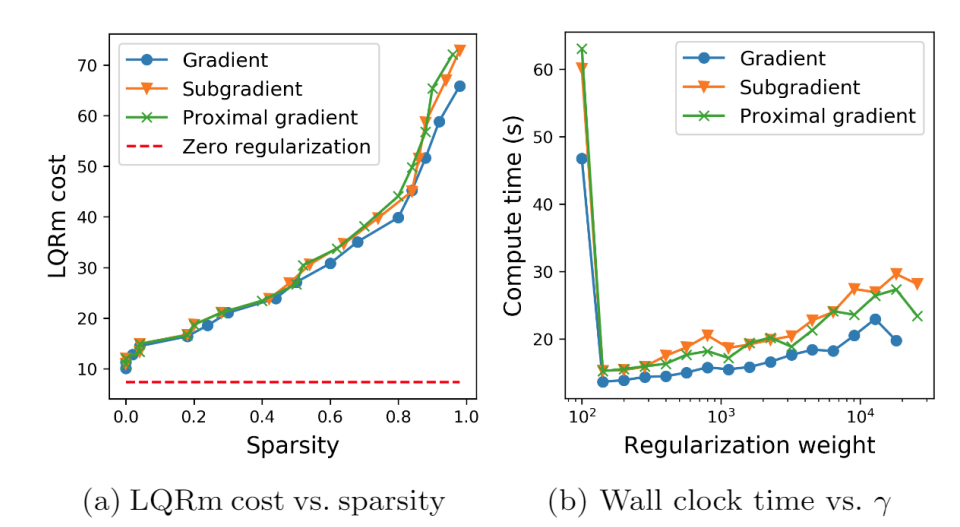


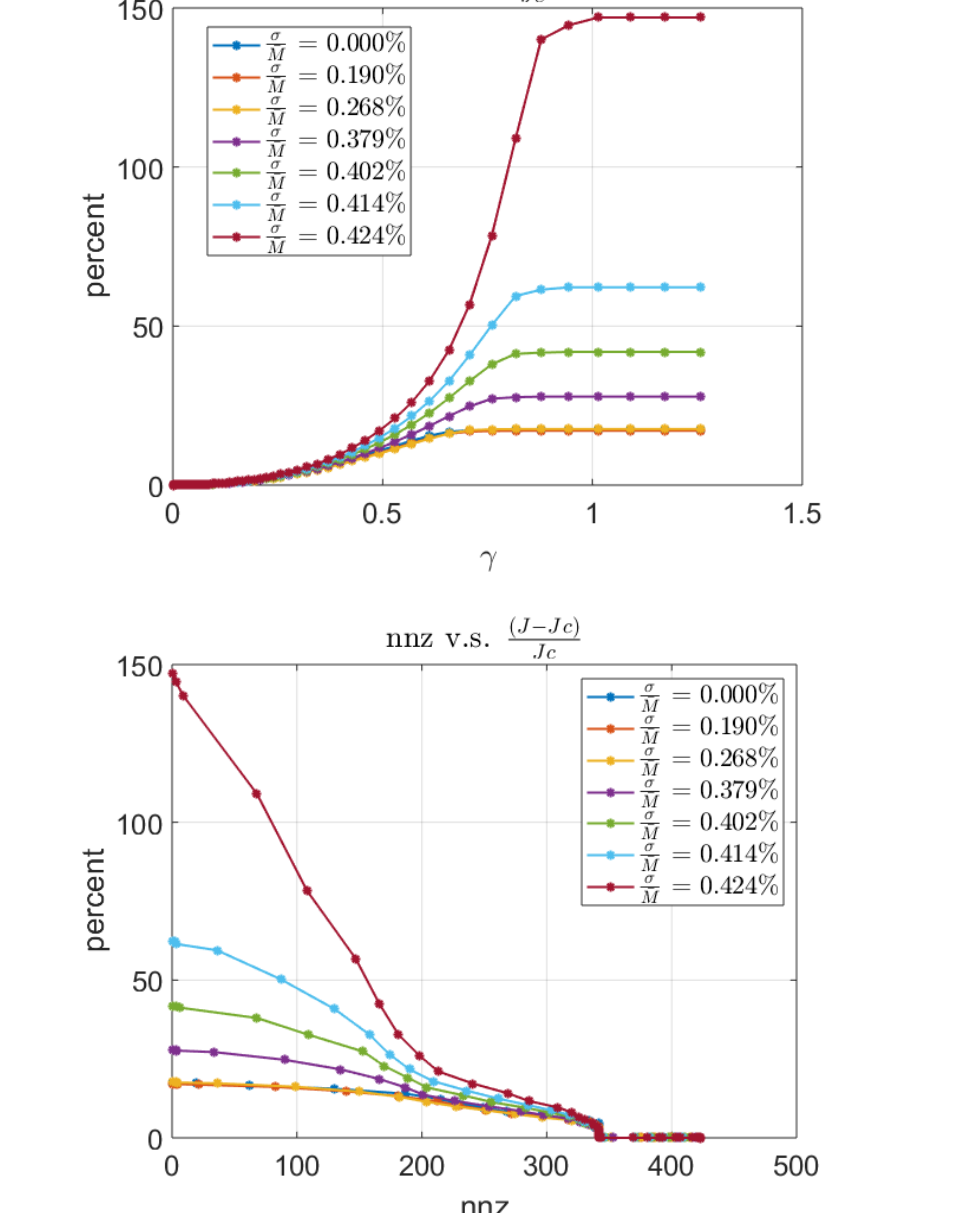
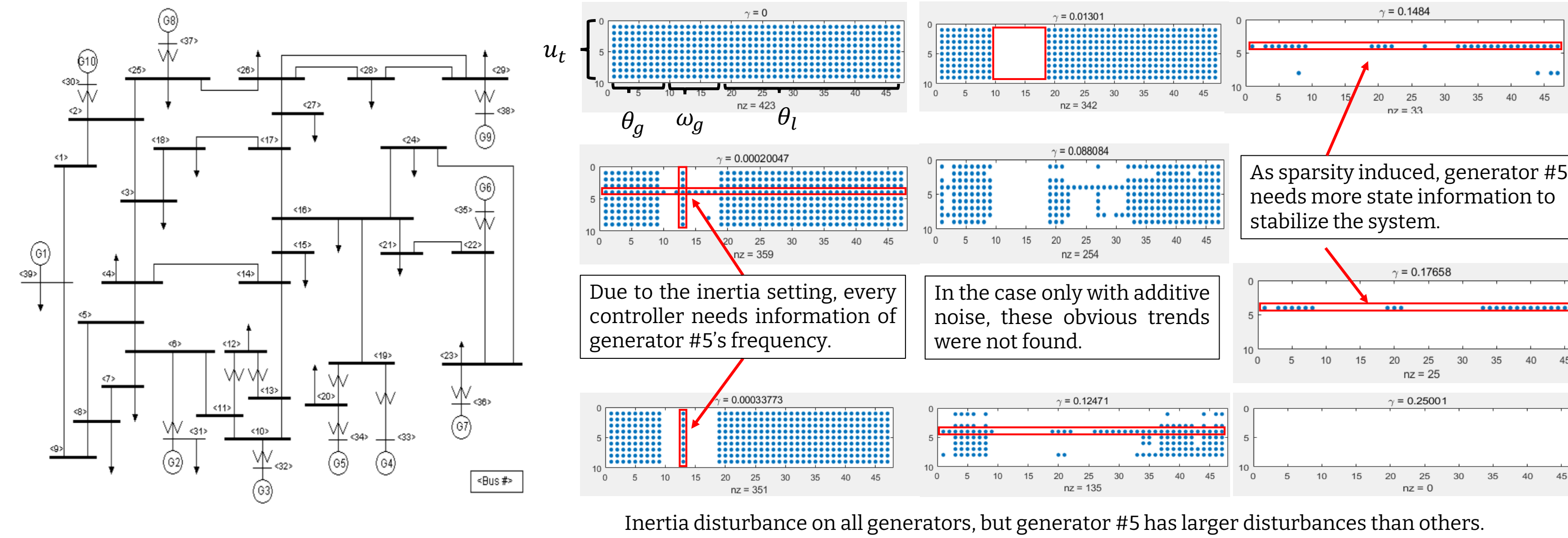
Fig. 4. Algorithm comparisons for the low noise setting with row-group LASSO regularization.

NUMERICAL EXPERIMENTS: LOW-INERTIA POWER GRIDS

IEEE 39-bus New England Testbed

Varying communication topologies under different sparsity levels

Performance vs. controller sparsity



CONCLUSIONS

Multiplicative noise complicates control design since mean-square instability of the state can occur due to noise-ignorant control design even when the mean is stabilized

Policy gradient provably finds optimal controllers in the unregularized model-free setting

Regularized policy gradient finds performant, sparse controllers for large networked systems and readily extends to the model-free setting

Future Research Directions

- Sufficient conditions and analysis of convergence of the sparse optimal control optimization
- Investigation of accelerated gradient descent schemes e.g. due to Nesterov, Polyak and second-derivative methods such as Newton and quasi-Newton
- Robustness guarantees under mean-square stability for the adaptive control setting, where parameter uncertainty must be addressed
- Gradient-based schemes for optimal estimation and sparse observer design

PAPER	https://arxiv.org/abs/1905.13548	
CODE	https://github.com/TSummersLab/polgrad-multinoise	
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